

Sikkandar Sulaiman A

Embedded Software Engineer

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projects | Git | LinkedIn Jurong, Singapore

7 years of product development experience primarily in Automotive industry using RTOS, embedded platforms and peripherals. Expertise in interfacing & programming hardware and design & developing production-level code. Enthusiast in designing clean modules, implementing with code readability and maintainability.

TECHNICAL SKILLS

C, C++, Python, OOAD	Linux, SafeRTOS, Bare Metal
ADAS, AUTOSAR, ASPICE	CAN, I2C, SPI, UART, sockets
ARM Cortex R, BSP, Crash Dump	HIL Debugging, Unit tests, Peer reviews
Git, Agile, IBM RTC, CodeBeamer	CI/CD pipeline, Jenkins, Coverity, VectorCAST
Sensors, Actuators, Micro Peripherals, Drivers	GUI dashboards and tasks automation
Full Stack, REST, JavaScript, SQL, Design Patterns	Lauterbach TRACE32, CANalyzer, MATLAB

PROFESSIONAL EXPERIENCE

Qualcomm Bangalore, India, Feb 2022 to June 2024

Product: BSP for Snapdragon Automotive chipsets

- Embedded firmware and BSP drivers development for **ARM Cortex R SOC**s in Automotive environment.
- Developed drivers adhering to static analysis, MISRA in SafeRTOS and AUTOSAR pro spaces.
- Implemented those drivers with crucial analysis in **pre-silicon FPGA and post-silicon board bring-up**.
- Contributed for memory dump parsers development to distill logs, call stack, registers and RAM info.
- **Reduced debugging time for multiple teams by creating data extraction plugins for parser tool using C#.**
- Worked in **multiple product lines and modules simultaneously** across fast-paced global teams.

Awak Technologies Bangalore, India, Oct 2020 to Jan 2022

Product: Automated Peritoneal Dialysis Medical Therapy device

- Designed & developed **multi-threaded firmware for delivering therapy** using RTX5 RTOS in STM32 micro.
- Drivers and algorithm for actuators, various feedback sensors including pressure, color, temperature etc.
- Developed heater module to maintain fluid temperature based on **closed-loop temperature feedback**.
- Designed & implemented a UART messaging protocol between backend therapy MCU and HMI tools.
- **Achieved testing time and resource reduction by creating a full-stack framework for debugging, logging.**
- Made the tool compatible with the messaging protocol and created real-time GUI using JavaScript.

Zilogic Systems (deployed in Visteon India) Chennai, India, Aug 2017 to Sep 2020

Product: Automobile Instrument Cluster for Hyundai OEM

- **Open-source contribution** such as Linux Test Project (LTP) and 5G (LDPC encoder bugfix).
- Designed & developed features including ADAS features, temperature unit, telltales, warnings, TPMS.
- Contributed to **core layers like CAN, RTE, MCAL**. Strong coding practices like unit tests, peer reviews.
- Worked closely with global cross-functional tech teams, from requirements, prototype to production.
- **Improved quality by simplifying error tracking for sprints by automating CI reports using Python scripts.**
- Extracted MISRA, Coverity, unit and functional test reports during CI build, made GUI and posted in cloud.

EDUCATION

MSc - Computer Control and Automation (Electrical and Electronic Engineering)

Nanyang Technological University, Singapore

2024-2025

BE - Electronics & Communication Engineering

Sri Shakthi Institute of Engineering & Technology, TN, INDIA

2013-2017

PUBLICATION

[Control of vehicle pollution through Internet of Things \(March 2017\)](#) ↗

[International Journal of Advance Research, Ideas and Innovations in Technology \(IJARIIT\)](#)

Successfully prototyped this as the major and final project work for the Bachelors program.

UNDERGRAD ACADEMIC PROJECTS

[Vehicle Pollution control using spectrophotometry for emission detection](#)

March 2017

- Demonstrated a minimal spectrophotometer setup, to detect harmful chemical compounds found in vehicle emission by analyzing the wavelength variation of light using a diffraction grating and a photocell.
- Used Raspberry Pi to compute and display values and send accumulated data to a monitoring server.
- Emission measurements are monitored in a regular basis and informed to Pollution Monitoring Center if no action is taken by the user to reduce emission levels for an aforesaid period of time.

[Home automation suite using Tele-robot](#)

May 2016

- Built a tele-robot, operated with Raspberry Pi and a camera attached. The video is live streamed over the internet, enabled by port forwarding.
- Created a web page to telecast the live video stream and made web UI controls to operate the robot and switch the home appliances.
- The robot carries Raspberry Pi which connects to internet through Wi-Fi for streaming and receiving movement control commands, and it operates the motors of the robot connected to it.
- Home appliances are connected to ESP8266 Wi-Fi nodes in local network, commanded through RaspberryPi.

[Self-balancing robot](#)

Jan 2016

- Attempted to build a 2-wheel self-balancing robot, powered by Arduino platform.
- PID control logic was modelled in MATLAB Simulink and deployed to Arduino.
- Feedback data for PID is gathered using IMU sensor through I2C bus and PWM is adjusted for motors.

INTERESTS

Community activities, Mentorship, Badminton, Hiking, Tamil Literature